other commenter stated that broker-dealers that offer algorithmic trading would not be able to utilize a competing consolidator due to the inherent latency of third party aggregation.[710]

As discussed more fully below,[711] the Commission acknowledges that, unlike self-aggregators, competing consolidators would need to transmit consolidated market data to their customers,[712] but does not believe that this would lead to the development of a two-tiered market. Latency sensitive customers of competing consolidators are likely to be co-located in the same data centers as their competing consolidators, so the transmission time between the servers of the competing consolidator and its customer will be exceedingly small. The Commission expects that market participants that elect to aggregate consolidated market data, whether competing consolidators or self-aggregators, will innovate and compete aggressively on the efficiency and cost-effectiveness of their aggregation technologies to attract and retain subscribers (in the case of competing consolidators) or to facilitate their trading strategies (in the case of self-aggregators). The Commission believes that the development and implementation of the technology to collect, consolidate, and generate consolidated market data will create opportunities for latency efficiencies that are of substantially greater magnitude than the transmission time between the server of a competing consolidator and its customer. Competing consolidators, for example, may benefit from economies of scale that allow them to offer a very

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710 See NBIM Letter at 4. This commenter, however, also stated that from an asset manager's perspective, the proposal would reduce its needs for direct feeds if there is a competitive consolidated tape offering. Id.

711 See infra Section III.D.2(d).

712 In the Proposing Release, the Commission noted that self-aggregators could have a minor latency advantage over market participants that use a competing consolidator for their consolidated market data. See Proposing Release, 85 FR at 16791.

low-latency product more cost effectively that an individual self-aggregator. In some cases, a competing consolidator may have a latency or cost advantage, and in others a self-aggregator may have such advantages.[713] Competition may also impact the efficiency of choices.[714] Therefore, the Commission does not believe that self-aggregators would necessarily have a systematic latency advantage over customers of competing consolidators.

6. Comments on the Potential Impact on Costs for Consolidated Market Data

Several commenters said that the proposed decentralized consolidation model would result in a reduction in the cost of consolidated market data.[715] One commenter stated that having multiple competing consolidators will reduce the prices of consolidated market data and proprietary market data feeds.[716] Several commenters stated that competition could bring down fees or the cost of consolidated market data.[717] One commenter said that competing forces

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713 Self-aggregators could have a cost advantage over market participants that receive consolidated market data from a competing consolidator because self-aggregators will not be required to compensate a competing consolidator for its services. A self-aggregator will of course incur expenses to generate consolidated market data including the costs of having the systems capability to collect, consolidate, and generate consolidated market data. It may use a vendor to establish connectivity to an SRO or to perform aggregation or other functions necessary for generating consolidated market data. As a result, any potential cost advantage of a self-aggregator over market participants that purchase consolidated market data from competing consolidators may not be significant.

714 See infra Section V.C.4(b)

715 See BestEx Research Letter at 4; DOJ Letter at 4, 5; Committee on Capital Markets Letter at 6; IntelligentCross Letter at 5; Better Markets Letter at 3; RBC Letter at 5–6; State Street Letter at 3; Fidelity Letter at 3, 9; Wellington Letter at 1; BlackRock Letter at 5; IEX Letter at 3.

716 See BestEx Research Letter at 4.

717 See DOJ Letter at 3–4 ("The Department agrees with the SEC's belief that 'by introducing competition and market forces into the collection, consolidation, and dissemination process, the decentralized consolidation model would help ensure that consolidated market data is delivered to market participants in a more timely, efficient

should help market participants to access market data in a cost-effective manner,[718] and another commenter said that competition would maintain fair prices.[719] A commenter stated that the proposal would constrain the cost of consolidated market data through competition among consolidators and the requirement that the fees charged to competing consolidators by the SROs be subject to approval.[720] Another commenter said the introduction of competing consolidators could impact aggregation and dissemination costs but stated that the proposal did not explain how competing consolidators would address exchange market data fees.[721]

However, other commenters expressed uncertainty about whether the proposed decentralized consolidation model would lower the cost of consolidated market data[722] or believed that the proposed model would increase costs.[723] One commenter stated that the proposal lacked proof that competing consolidators would reduce market data costs, explaining that the proposal lacked sufficient guidance and analysis of how market data fees would be determined, how to define reasonable fees, and how the proposed model would control costs to

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and cost effective manner than the current centralized consolidation model.'"); Committee on Capital Markets Regulation Letter at 6; IntelligentCross Letter at 5; Fidelity Letter at 9; Wellington Letter at 1; ICI Letter at 4.

718 See Better Markets Letter at 3.

719 See RBC Letter at 5–6.

720 See IEX Letter at 3.

721 See Citadel Letter at 5.

722 See STANY Letter II at 5; Data Boiler Letter I at 46–47; TD Ameritrade Letter at 12; NYSE Letter II at 9.

723 See Cboe Letter at 23–24; FINRA Letter at 1, 2, 3, 4; Angel Letter at 21, 24; Kubitz Letter at 1; Nasdaq Letter IV at 23, 26, 47–48, 60; TD Ameritrade Letter at 15.

participants.[724] Another commenter said that competing consolidators would not result in enough competition to lower the cost of consolidated market data,[725] and a commenter stated that there was no guarantee that competition would improve costs.[726]

Commenters also stated that the proposed decentralized consolidation model would increase the cost of consolidated market data.[727] One commenter stated that nothing in the proposal supported the conclusion that competing consolidators would price consolidated market data economically efficiently.[728] This commenter argued that market data costs would be higher because differentiation would result in higher prices as differentiated competing consolidators would have fewer customers over which to spread their fixed costs.[729] This commenter also said that the proposed increased content of consolidated market data could increase costs and burden retail investors who have no need for the more comprehensive data.[730] The commenter also said that costs will be dependent on the effective national market system plan(s) and fee proposals.[731] Another commenter said that retail investors could incur "exponentially more" costs as a result of the proposal.[732]

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724 See STANY Letter II at 5. Similarly, another commenter said the proposal lacked a "reasoned analysis of expected costs and fees for market data under the decentralized consolidation model." NYSE Letter II at 9. See also id. at 19–20.

725 See Data Boiler Letter I at 46–47.

726 See TD Ameritrade Letter at 12.

727 See Nasdaq Letter IV at 23, 26, 47–48, 60; TD Ameritrade Letter at 15.

728 See Nasdaq Letter IV at 23.

729 See id. at 26.

730 See id. at 60, n.162.

731 See id. at 47–48.

732 TD Ameritrade Letter at 15.

One commenter stated that product differentiation among competing consolidators could lead to an increase in prices as the fastest competing consolidators would charge more due to inelastic demand.[733] This commenter also believed that only the existing SIPs could afford Regulation SCI compliance; therefore, as the only competing consolidators, they would charge oligopolistic prices for consolidated market data.[734] This commenter also said that the real prices for consolidated market data are determined by what the Commission will permit the exchanges to charge, not competition.[735]

Several commenters stated that the proposed decentralized consolidation model would result in higher consolidated market data costs[736] as well as other costs[737] for market participants. One commenter said that competing consolidators would "impose meaningful costs on investors" and said that the proposal did not explain how competing consolidators would charge fees to investors (such as whether they could charge additional fees for content or only for data delivery).[738] This commenter also said that the proposal is deficient because ambiguities surrounding fees to be charged by competing consolidators to investors, as well as by SROs to competing consolidators, impede meaningful public comment and the ability of the Commission

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733 See Angel Letter at 23.

734 See id. at 20.

735 See id. at 21.

736 See Cboe Letter at 23–24; Kubitz Letter at 1; Angel Letter at 21, 24; Nasdaq Letter IV at 27, 30, 60, n.162; Data Boiler Letter II at 1.

737 See FINRA Letter at 4; Nasdaq Letter IV at 5, 27, 30.

738 Cboe Letter at 23–24.

to perform a cost-benefit analysis as required under the APA.[739] One commenter said that exchanges facing competitive pressure from shareholders will be forced to charge high prices to competing consolidators, which will then pass down these prices to their subscribers.[740] Another commenter said that competing consolidators are "an intermediary between suppliers and users adding a layer of cost to the overall system."[741]

Two commenters stated that market participants would face increases in other costs as a result of the proposal.[742] One of the commenters said that exchange trading costs for retail and other investors will increase due to reductions in SROs' market data revenue as a result of the proposal.[743] The other commenter said that costs would increase as a result of requiring broker-dealers (or other market participants) to subscribe and pay fees to multiple competing consolidators.[744]

The Commission recognizes that the fees for the data content underlying consolidated market data are unknown at this time and that such fees are a fixed cost for all competing consolidators to assess when developing their business plans. However, in response to the comments that expressed uncertainty about the direction of consolidated market data costs as a result of the proposal and those comments that stated that consolidated market data costs would

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739 See Cboe Letter at 4.

740 See Angel Letter at 24.

741 Data Boiler Letter II at 1.

742 See FINRA Letter at 4; Nasdaq Letter IV at 5, 27, 30.

743 See Nasdaq Letter IV at 5, 27, 30. See also Clearpool Letter at 3 (suggesting safeguards to keep exchanges from increasing consolidated market data prices to recoup any revenue lost from the proposed requirement to sell core data to competing consolidators).

744 See FINRA Letter at 4.

increase for market participants, the Commission believes that competition will constrain the prices at which competing consolidators can sell consolidated market data products. Further, the Commission believes that there will be downward pressure on the fees for the data content underlying consolidated market data as compared to fees for proprietary data.[745]

Specifically, the new data content underlying consolidated market data (i.e., depth of book data, auction information, and odd-lot information) are currently elements of proprietary data products that are assessed under the statutory standards that apply to proprietary data, including Sections 6(b)(4), 15A(b)(5), and 11A(c)(1)(C)-(D) of the Exchange Act[746] and Rule 603(a) under Regulation NMS.[747] These proprietary data fees are filed with the Commission pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder[748] and are effective upon filing with the Commission.

Fees for the data content underlying consolidated market data will be assessed against the statutory standard that applies to fees proposed by the effective national market system plan(s), including Sections 11A(c)(1)(C) – (D) of the Exchange Act and Rule 603(a) under Regulation NMS. The proposed fees must be fair and reasonable and not unfairly discriminatory.[749] The fees must be filed with the Commission pursuant to Rule 608 and will be published for public

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745 See infra Section III.E.2(c) (discussing the statutory requirements applicable to consolidated market data and the standards the Commission has historically applied to assessing compliance with the statutory requirements).

746 15 U.S.C. 78f(b)(4), 15 U.S.C. 78*o*-3(b)(5), and 15 U.S.C. 78k-1(c)(1)(C)-(D).

747 17 CFR 242.603(a).

748 17 CFR 240.19b-4.

749 See infra Section III.E.2(c).

comment and thereafter, if consistent with the Exchange Act, must be approved by the Commission before becoming effective.[750]

In addition, a New Consolidated Data Plan has been filed that contains a proposed new governance structure and procedures that, if approved, will address some of the conflicts of interest inherent in the existing governance structure and will bring a more inclusive representation of market participants into the process for developing fees for the data content underlying consolidated market data.[751] The Commission believes that the governance model required to be included in the proposed New Consolidated Data Plan will support the building of broad consensus in developing the future fees for the data content underlying consolidated market data. Notwithstanding the new governance model, the new fees for data content underlying consolidated market data will have to satisfy statutory standards: they must be fair and reasonable and not unfairly discriminatory.[752]

Further, competition should constrain other aspects of consolidated market data costs, including the fees charged by competing consolidators for their products and services. Competing consolidators will compete in their aggregation and transmission services, which should also be reflected in their prices for such services. All competing consolidators will be required, under Rule 614(d), to disclose publicly metrics and other information concerning their performance and operations, which will allow market participants to evaluate effectively

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750 See Effective-Upon-Filing Adopting Release supra note 17.

751 See Governance Order, infra note 1128; New Consolidated Data Plan Notice, supra note 40.

752 See Sections 11A(c)(1)(C)-(D) of the Exchange Act and Rule 603(a) of Regulation NMS, 17 CFR 242.603(a).

competing consolidators, fostering competition among competing consolidators. Further, competing consolidators are required to disclose their prices for consolidated market data products.[753] In response to the comments warning that product differentiation would permit competing consolidators to increase their fees for consolidated market data products, the Commission believes that competing consolidators will face competition from each other and from potential new entrants, especially if a differentiated product serves as a material economic opportunity. This threat of competition will discipline prices and efficiency in the consolidated market data space. For example, if products tailored to the different needs of market participants become popular, competition should drive the creation and sale of similar products at prices attractive to subscribers.

In response to the comment that stated that exchange trading costs, and consequently retail investor trading costs, would increase due to a reduction in exchange revenue as a result of the proposal,[754] the Commission notes that the exchanges may file proposed rule changes to reflect any necessary adjustments to their fees as a result of the proposal. These proposed rule changes must meet the applicable statutory standards for fees. However, a reduction in proprietary data revenue, if it were to occur, would not, by itself, necessarily make it optimal for exchanges to adjust their trading fees.[755]

In response to the comment that stated that market participant costs will increase as a result of the proposal because all market participants would need to retain a back-up competing consolidator,[756] the Commission is not requiring market participants to have back-up competing

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753 See Form CC, Exhibit F.

754 See supra note 743.

755 See note 2392 and accompanying text.

756 See FINRA Letter at 4.

consolidators. Market participants may choose to subscribe to competing consolidators that are "SCI competing consolidators" — those subject to the requirements of Regulation SCI that help ensure that the core technology systems of SCI entities remain reliable and resilient, including the requirements to have geographically diverse back-up and recovery capabilities, and conduct an SCI review each year, as discussed below. Some market participants may choose to subscribe to multiple competing consolidators. In either case, this choice will be for market participants to elect after evaluating the needs of their business and their customers. The Commission cannot estimate at this point specific cost increases, if any, for market participants that subscribe to competing consolidators.[757] But market participants may face an overall reduction in costs due to the competitive environment for consolidated market data fostered by the decentralized consolidation model.

7. Comments on Complexity of the Decentralized Consolidation Model

Several commenters stated that the proposed decentralized consolidation model would generally add complexity to the consolidated market data environment.[758] One commenter said that competing consolidators would increase technological complexity, which would also

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757 Those broker-dealers that do not currently have back-up capabilities may decide not to have back-up capabilities under the decentralized consolidation model. To the extent that such broker-dealers decide to subscribe to redundant back-up competing consolidator feeds, they may incur higher costs. Further, as discussed below, the effective national market system plan(s) will have to develop MISU policies for consolidated market data subscribers. See infra Section III.E.

758 See TechNet Letter II at 1–2; STANY Letter II at 8.